

A preliminary evaluation of wave energy resource in YYYY

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ABSTRACT: This study evaluates the **wave energy resource** along the coastal regions of YYYY using long-term directional wave spectra from offshore buoy measurements. More than a decade of continuous records ensures statistical robustness for characterizing the regional wave climate. After rigorous quality control, the offshore wave conditions were **numerically transformed** toward selected port locations to account for nearshore bathymetry and coastal geometry. The resulting time series were analysed using univariate, bivariate, and directional statistical methods to quantify spatial and seasonal variability. Wave power was then estimated through spectral energy formulations and probability scatter tables representative of the study period. The results provide a high-resolution characterization of the **exploitable wave energy potential** along the YYYY coastline, supporting future resource mapping and site-screening efforts.

Keywords: netCDF, statistical analysis. variability, wave energy resource

Table 6.1 – Mean annual power [Kw/m]

2012	Winter		Summer		Autumn		Spring		Annual	
γ	3.3	1	3.3	1	3.3	1	3.3	1	3.3	1
TS_1	33.17	31.40	31.32	29.64	32.04	30.33	25.16	23.82	24.62	23.30

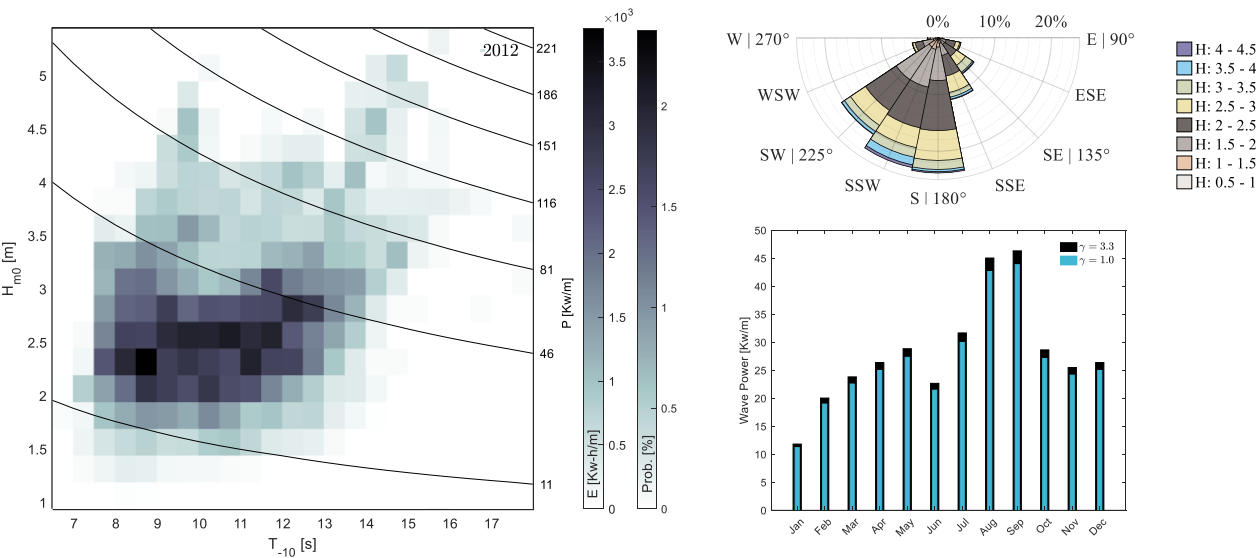


Figure - (left) Scatter plot (right-bottom) monthly wave energy flux distribution (right-top) wave heights rose diagram to the year 2012